ML Miniproject

Problem 11: Application of Artificial Neural Network in Streamflow Forecasting

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1. Problem Statement

Streamflow forecasts are vital for flood early warning, reservoir management, and water-resource planning. We formulate one-day-ahead daily streamflow prediction as a supervised nonlinear autoregressive regression problem with exogenous rainfall inputs:

$$Q(t) = f\big(R(t-1), \dots, R(t-m),\ Q(t-1), \dots, Q(t-n)\big)$$

where Q denotes streamflow discharge, R denotes precipitation, and m, n are the respective input lag horizons.

Our objective is to evaluate a feed-forward Artificial Neural Network (ANN) against standard persistence and linear baselines, with targeted evaluation on extreme/peak-flow regimes. The experimental premise adapts the Leaf River formulation from Xiao & You (Stanford CS229, 2020). While adopting their station and target setup, we employ an independent dataset pipeline, distinct temporal splits, multi-seed validation, and separate evaluation protocols.

2. Dataset & Preprocessing

- Station & Variables: Leaf River near Collins, Mississippi (USGS 02472000; drainage area: 743 sq mi). Daily mean discharge is extracted in cubic feet per second (cfs) via USGS NWIS. Daily precipitation (mm/day) is sourced from Daymet v4 single-pixel gridded estimates at the gauge coordinates.
- Temporal Period & Split: January 1, 2003 to December 31, 2012 (3,653 calendar days). We apply a strict chronological, non-shuffled split:
  - Training: 2003–2006 (4 years)
  - Validation: 2007 (1 year, reserved solely for model selection and early stopping)
  - Test: 2008–2012 (5 years, 1,824 daily steps; evaluated once)
- Feature Engineering & Leak Prevention: Dates were verified against a daily calendar. Gapped windows were excluded without artificial interpolation. Predictors strictly use past days ($t-1$ backwards); same-day rainfall is excluded to prevent look-ahead bias. Both rainfall and streamflow series are normalized using training-set maxima only, preventing data leakage into validation/test splits.

3. Methodology & Architecture

- Model Configuration: Fully connected Multi-Layer Perceptron (MLP) featuring one or two hidden layers with sigmoid activations (matching reference specifications) and a single linear output unit for continuous flow estimation. Optimization uses Adam on Mean Squared Error (learning rate: 0.005, batch size: 32, max epochs: 200). Training applies early stopping with a 25-epoch patience window on validation MSE, restoring best validation weights.
- Hyperparameter Grid Search: Evaluated 36 candidate configurations across $m \in \{2,3,5\}$, $n \in \{2,4,6\}$, and hidden topologies $\in \{(8), (12), (16), (16,8)\}$ using 3 random seeds per candidate on 2007 validation RMSE. The optimal setup—$m = 3$, $n = 2$, hidden layers $(16,8)$ (validation RMSE: 281.9 ± 8.2 cfs)—was retrained across 5 random seeds to compute mean and standard deviation.

- Baselines & Metrics:
    1. *Persistence Baseline:* $\hat{Q}(t) = Q(t-1)$.
    2. *Multiple Linear Regression:* Fitted over identical lag horizons ($m = 3, n = 2$).
    - *Metrics:* Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Nash–Sutcliffe Efficiency ($NSE/R^2 = 1 - \frac{\text{SSE}}{\text{SST}}$).
    - *Regime Stratification:* Test samples are bifurcated into peak days ($Q(t) > 1500$ cfs; $N = 225$, 12.3% of test samples) and non-peak days ($Q(t) \leq 1500$ cfs).

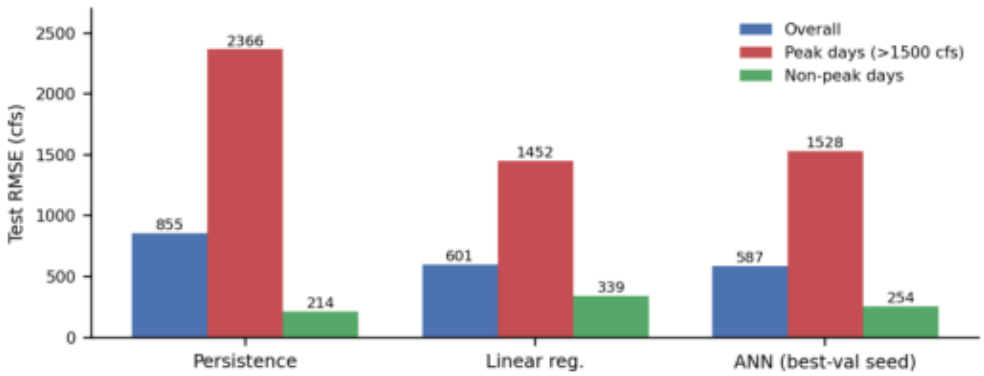
4. Implementation Details

The framework is implemented in Python 3 using pandas, NumPy, scikit-learn, and matplotlib. Neural models utilize MLPRegressor driven epoch-by-epoch via partial_fit to record training/validation trajectories and govern patience-based early stopping. The repository includes modular components: download_data.py (API extraction), src/ modules (feature lag generation, scaling, architecture, and visualization), pipeline.py (end-to-end execution), and an interactive demonstration notebook.

5. Results & Discussion

Table 1: Model performance on the unseen test partition (2008–2012). All flow metrics in cfs.

Model	MAE (cfs)	RMSE (cfs)	R2 / NSE	RMSE (Peak >1500)	RMSE (Non-Peak)
Persistence Baseline $Q(t-1)$	278.9	854.8	0.768	2366	213.8
Linear Regression	291.5	600.8	0.886	1452	339.5
ANN (Best Validation Seed)	237.7	587.0	0.891	1528	253.8
ANN (5-Seed Ensemble: Mean $\pm$ SD)	272.0 $\pm$ 22.6	587.2 $\pm$ 10.9	0.891 $\pm$ 0.004	1495 $\pm$ 65	278.9 $\pm$ 28.0



- ANN vs. Baselines: The ANN achieves a 31.3% reduction in overall RMSE compared to persistence (587.2 vs. 854.8 cfs) and improves NSE from 0.768 to 0.891. However, relative to multiple linear regression (RMSE: 600.8 cfs), the ANN shows only a marginal ~2.2% gain, roughly within one seed standard deviation (10.9 cfs). We therefore do not claim statistical superiority over linear models for this feature set.
- Peak-Flow Degradation: Forecasting errors are heavily concentrated during high-flow episodes (ANN peak RMSE: 1495 cfs vs. non-peak RMSE: 279 cfs). Linear regression slightly outperforms

the ANN on peak events (1452 vs. 1528 cfs). Conversely, simple persistence proves most effective for baseflow regimes (non-peak RMSE: 213.8 cfs), while the ANN minimizes non-peak absolute error (MAE: 237.7 cfs).

- Validation Shift & Reference Discrepancies: The selected model scored 281.9 cfs on the 2007 validation set but rose to 587.0 cfs on the test partition, showing that a single validation year is sensitive to hydrologic variability. Xiao & You reported a normalized ANN RMSE of 0.23% of peak flow, whereas our best model achieved 3.14%. This divergence stems from differing rainfall products (point gauge vs. single-pixel Daymet) and out-of-sample data splits.

6. Conclusions & Limitations

A compact feed-forward ANN using lagged antecedent rainfall and discharge outperforms naive persistence by over 31% in overall RMSE, but yields marginal improvements over linear regression and fails to alleviate extreme peak-flow underestimation.

Key Limitations:

1. Evaluation is restricted to a single watershed using single-pixel Daymet gridded rainfall rather than distributed gauge networks.
2. Standard MSE loss penalizes all discharge errors equally, causing models to favor high-frequency baseflow at the expense of rare flood peaks.
3. Hyperparameter search resolution was constrained, and single-year validation does not encompass decadal hydrological variance.

Future Work: Incorporating same-day radar precipitation, testing peak-weighted/asymmetric loss functions, applying logarithmic discharge transformations, and implementing Recurrent/LSTM networks to capture multi-day catchment retention dynamics.

References

1. Xiao, C., & You, A. (2020). *Application of Artificial Neural Network in Streamflow Forecasting*. Stanford CS229 Project Report.
2. Thornton, M. M., et al. (2020). *Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4*. ORNL DAAC.
3. U.S. Geological Survey (USGS). *National Water Information System (NWIS)*: Leaf River near Collins, MS (Gauge 02472000).